Options II: Properties

BUSS386. Futures and Options

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Lecture Outline

- Why discounted cash flow cannot price an option — and what we can do instead
- The six factors that move an option's price, and what changes once early exercise is allowed
- How far no-arbitrage alone pins down a price: upper and lower bounds
- **Put-call parity** — the single identity tying c, p, S and K together
- When early exercise is rational, and why a call on a non-dividend stock never is
- What changes once the stock pays dividends, discrete or continuous

Factors Affecting Option Prices

Why DCF fails for options

The natural first attempt. From corporate finance, value anything by discounting its expected payoff. For a European call,

$$c_0 = \underbrace{e^{-r_{\text{call}}\,T}}_{\text{discount factor}} \times \underbrace{E\big[\max(S_T - K,\, 0)\big]}_{\text{expected payoff}}.$$

- **The expectation:** It needs a model for the distribution of S_T.
- **The discount:** r_{call} is the return investors require on *the option*, and an option is a leveraged claim on the stock: its risk — and so r_{call} — changes continuously as S moves.
- Before building any model, ask what must be true of c_0 and p_0 purely to rule out arbitrage. That costs no assumptions, and it still pins the price down a long way: **bounds** and **put-call parity**.

The six factors

Factor	Eur. c	Eur. p	Am. C	Am. P
Spot S_0				
Strike K				
Volatility σ				
Dividends D				
Risk-free rate r				
Time to expiry T				

- Time to expiry: A longer-dated American option carries every exercise opportunity the shorter one has, and more, so more time can never hurt. A European holder has only the date T itself, so a longer wait cuts both ways — a large dividend may fall inside the new window, and on a put you simply receive K later.

Factors Affecting Option Prices

Q. Consider two European put options with the same strike prices and the same expiration dates but for different underlying stocks A and B. Stock A has the volatility of 20% and is expected to pay dividend \$3 in a year. The stock B has the volatility of 15% and is expected to pay dividend \$2 in a year. The current stock prices of A and B are the same. Can we determine which put option has the higher price?

Lower/Upper Bounds for Option Prices

Put-Call Parity

Why bounds and parity, instead of a price

- To exactly determine the option price, we need a model that describes futures stock price.
- We don't want to use a model yet.
- Instead, we use the no-arbitrage argument only and try to characterize the option price as much as possible.
- This results in ...
  - Lower/upper bound for the option price
  - Put-call parity

European vs American Options

- Consider an American option and a European option with the same strike prices, expiration dates, and underlying assets.
- The American option is always more valuable or as valuable as the European option.

$$c_0 \leq C_0 \quad \text{and} \quad p_0 \leq P_0$$

- This is because the owner of the American option has all exercise opportunities open to the owner of the European option and more.
- This relationships hold for all types of options irrespective of whether the underlying asset pays dividends or not.

The four cases we work through

- We consider lower/upper bounds and the put-call parity case by case.

1. Non-dividend-paying stock
   - European options
   - American options

2. Dividend-paying stock
   - Continuous dividend
     - European options
     - American options
   - Discrete dividend
     - European options
     - American options

Lower/Upper Bounds for Option Prices

Put-Call Parity

I. Options on Non-Dividend-Paying Stocks

Upper Bounds — Non-Dividend Stock

Calls. The option to buy S at K is never worth more than S itself (buying the stock outright is at least as good):

$$c_0 \le S_0, \qquad C_0 \le S_0.$$

Puts. European put pays at most K at T. Discounting:

$$p_0 \le K\, e^{-rT}.$$

American put can be exercised any time, so its discounted max payoff is just K:

$$P_0 \le K.$$

- These are crude but *always* hold; useful sanity checks on quotes.

Lower Bound — European Call (Non-Dividend)

$$c_0 \;\geq\; \max\bigl(S_0 - K\,e^{-rT},\; 0\bigr).$$

Proof sketch. Compare two portfolios:

Portfolio	Today	At T
(A) Long call + bond paying K at T	$c_0 + Ke^{-rT}$	$\max(S_T, K)$
(B) Long one share	S_0	S_T

At T, A's payoff $\geq$ B's payoff (since $\max(S_T, K) \geq S_T$). By no-arbitrage:

$$c_0 + Ke^{-rT} \;\geq\; S_0 \quad\Longrightarrow\quad c_0 \;\geq\; S_0 - Ke^{-rT}.$$

Combining with $c_0 \geq 0$ gives the boxed bound.

Example — Call Lower-Bound Arbitrage

Q. $S_0 = \$20$, $K = \$18$, $r = 10\%$, $T = 1$. A call is quoted at \$3. Arbitrage?

- **A.** Lower bound $= \max(20 - 18e^{-0.1}, 0) = \max(20 - 16.287, 0) = \boxed{\$3.713}$.
- Call price \$3 $<$ \$3.713 $\Rightarrow$ **arbitrage**: buy the call, short the stock, lend Ke^{-rT}:

Action today	CF now	CF at T	
		$S_T \geq 18$	$S_T < 18$
Buy call	-3	$S_T - 18$	0
Lend $18e^{-0.1}$	-16.287	18	18
Short share	$+20$	$-S_T$	$-S_T$
Net	$\boxed{+0.713}$	0	$18 - S_T \geq 0$

Lower Bound — European Put (Non-Dividend)

$$p_0 \;\geq\; \max\bigl(K\,e^{-rT} - S_0,\; 0\bigr).$$

Proof sketch. Compare:

Portfolio	Today	At T
(C) Long put + one share	$p_0 + S_0$	$\max(S_T, K)$
(D) Long bond paying K at T	Ke^{-rT}	K

At T, C $\geq$ D (since $\max(S_T, K) \geq K$):

$$p_0 + S_0 \;\geq\; Ke^{-rT} \quad\Longrightarrow\quad p_0 \;\geq\; Ke^{-rT} - S_0.$$

Put-Call Parity - Derivation 1

[European Call and Put]

$$\boxed{c_0 + K\, e^{-rT} \;=\; p_0 + S_0}$$

- Recall the two portfolios used for the bounds:
  - (A) European call + bond that will pay K at T
  - (C) European put + one share
- At expiry T, (A) and (C) always generate the *same* cash flows:
  - (A) $\max(S_T - K, 0) + K = \max(S_T, K)$
  - (C) $\max(K - S_T, 0) + S_T = \max(K, S_T)$
- Hence, under no-arbitrage, current prices should satisfy

$$c_0 + Ke^{-rT} = p_0 + S_0$$

Put-Call Parity - Derivation 2

[European Call and Put]

$$\boxed{c_0 - p_0 \;=\; (F_0 - K)\, e^{-rT}}$$

- Consider a portfolio of options: + long Call + short Put
- The payoff of this portfolio at T is $S_T - K$, which is the same as the payoff of a forward with *delivery price* K — not a forward whose forward price is K.
- The present value of that payoff is $S_0 - Ke^{-rT}$.
- Therefore, $c_0 - p_0 = S_0 - Ke^{-rT}$.

Forward-Style Parity — One Formula Covers Everything

For any underlying with forward price F_0 — stocks with dividends, FX, futures, indices:

$$\boxed{c_0 - p_0 \;=\; (F_0 - K)\, e^{-rT}}$$

- **Non-dividend stock.** $F_0 = S_0 e^{rT} \Rightarrow c - p = S_0 - Ke^{-rT}$ — the form we just derived.
- **Continuous dividend yield q.** $F_0 = S_0 e^{(r-q)T} \Rightarrow c - p = S_0 e^{-qT} - Ke^{-rT}$.
- **Discrete dividends, present value D.** $F_0 = (S_0 - D)e^{rT} \Rightarrow$ $c - p = (S_0 - D) - Ke^{-rT}$.
- **FX, foreign rate r_f.** $F_0 = S_0 e^{(r-r_f)T} \Rightarrow c - p = S_0 e^{-r_f T} - Ke^{-rT}$.

Takeaway. Learn parity in this one form. Every version you meet later is the same identity with a different F_0 substituted in.

Example — Arbitrage with Parity

[European Call and Put]

Q. Suppose that $S_0 = \$31$, $K = \$30$, $r = 10\%$, $T = 3$ month. The price of a European call is \$3 and the price of a European put is \$2.25. Is there an arbitrage opportunity? If so, find an arbitrage strategy and its profit.

Answer: Let's check whether the put-call parity holds:

1. $c_0 + Ke^{-rT} = 3 + 30e^{-0.1\times 3/12} = 32.26$
3. $p_0 + S_0 = 2.25 + 31 = 33.25$

So, the portfolio ③ is overpriced relative to portfolio ①. Then, the arbitrage strategy is

Action today	Today	T	
		$S_T \geq 30$	$S_T < 30$
long call	-3	$(S_T - 30)$	0
buy a bond	$-30e^{-0.1\times 3/12}$	30	30
short put	2.25	0	$-(30 - S_T)$
sell share	31	$-S_T$	$-S_T$
net	0.99	0	0

American Options - Early Exercise (1 of 4): the question

[Non-dividend-paying stock]

- Long position in an American option has the right to exercise earlier than the expiration.
- In a special case, when we long an **American call on a non-dividend-paying stock**, it is **never optimal to exercise early** before the expiration (for reason we will see below).
- Why is early exercise not optimal for this special case?
  - At time t, the option holder has the value

$$C_t = \max \begin{cases} \text{Value of waiting} & \text{if not exercise} \\ S_t - K & \text{if exercise} \end{cases}$$

American Options - Early Exercise (2 of 4): via put-call parity

[Non-dividend-paying stock]

- If $C_t > S_t - K$, we can conclude that early exercise is not optimal.
- It turns out that the above inequality holds at any $0 \leq t < T$.
  - As an American call option is worth more or as much as a European call,

$$C_t \geq c_t$$

  - From the put-call parity of European options, we have

$$\begin{aligned} c_t &= p_t + S_t - Ke^{-r(T-t)} \\ &= p_t + (S_t - K) + K(1 - e^{-r(T-t)}). \end{aligned}$$

    Thus, $c_t > S_t - K$.
  - Hence, $C_t > S_t - K$. It's better to sell the call.

American Options - Early Exercise (3 of 4): via dominance

[Non-dividend-paying stock]

- Alternatively ...
- Consider the following two portfolios:
  1. American call
  2. One share + short bond that will pay K at T
- At the option expiration T, ① always generates larger cash flows than ②:
  1. $\max(S_T - K, 0)$
  2. $S_T - K$
- Hence, under no-arbitrage, at $t < T$,

$$C_t \geq S_t - Ke^{-r(T-t)} \geq S_t - K$$

American Options - Early Exercise (4 of 4): why puts are different

[Non-dividend-paying stock]

- In the case of an American **put** option, sometimes it is optimal to exercise early.
  - $P_t \leq K$ because the maximum payoff from the option is $K - 0$.
  - Suppose $S_t = 0$, the payoff at t is K and we know $P_t \leq K$.
  - It's better to receive K earlier than later.
  - Hence, it can be desirable to exercise the option at $t \leq T$.

Lower Bounds for American Call

[Non-dividend-paying stock]

- We just proved that the option expiration is the only date that we may exercise an American call on non-dividend-paying stock.
- This means that the European and the American calls will deliver the same cash flows. Thus, $C_0 = c_0$.
- Hence, the lower bound of American call is the same as the lower bound of European call.

Lower Bounds for American Put

[Non-dividend-paying stock]

- For American put option, it is sometimes optimal to exercise early, in particular, when the option is deep in the money.

- At time t, the option holder has the value

$$P_t = \max \begin{cases} \text{Value of waiting} & \text{if not exercise} \\ K - S_t & \text{if exercise} \end{cases},$$

  Thus, we know $P_t \geq K - S_t$.

- Combining the fact that the put option price cannot be negative, the lower bound becomes

$$P_0 \geq \max(K - S_0, 0).$$

Put-Call Parity for American Options

- For American options on non-dividend-paying stocks, the put-call parity is

$$S_0 - K \leq C_0 - P_0 \leq S_0 - Ke^{-rT}$$

- Let's prove the right inequality first and then prove the left inequality.

Put-Call Parity for American Options - Right Inequality

- As $P_0 \geq p_0$, it follows that $C_0 - P_0 \leq C_0 - p_0$. Also, we know that $C_0 = c_0$ for non-dividend-paying stock. Thus,

$$C_0 - P_0 \leq C_0 - p_0 = c_0 - p_0$$

From the put-call parity for European options, we know that $c_0 - p_0 = S_0 - Ke^{-rT}$. Thus,

$$C_0 - P_0 \leq c_0 - p_0 = S_0 - Ke^{-rT},$$

which proves the right inequality.

Put-Call Parity for American Options - Left Inequality: setup

- To prove the left inequality, we consider the following two portfolios:

portfolio A:	American call + bond worth K now
portfolio B:	American put + stock

- We want to prove that the value of the portfolio A is higher than or equal to the value of portfolio B. This will lead to the left inequality, $C_0 - P_0 \geq S_0 - K$.

- In derivation, we consider the two different cases:

  1. case 1: put option is exercised earlier than the expiration.

  2. case 2: put option is not early-exercised.

Put-Call Parity for American Options - Left Inequality: case 1, put exercised early

- Suppose that the put option is exercised earlier than the expiration, say t $(0 \leq t < T)$.
- Then, the portfolio value at time t is

$$\begin{array}{ll} \text{portfolio A:} & C_t + Ke^{rt} \\ \text{portfolio B:} & (K - S_t) + S_t \end{array}$$

- As $C_t \geq 0$ and $e^{rt} \geq 1$, we can conclude that the time-t value of portfolio A is higher than or equal to the value of portfolio B.

Put-Call Parity for American Options - Left Inequality: case 2, held to expiry

- Suppose that the put option is NOT exercised earlier than the expiration. Then, the put option may or may not be exercised on the expiration T.

- If $S_T \geq K$ on the expiration T,

$$\begin{array}{ll} \text{portfolio A:} & (S_T - K) + Ke^{rT} \\ \text{portfolio B:} & 0 + S_T \end{array}$$

  Thus, the portfolio A has the higher value.

- If $S_T < K$ on the expiration T,

$$\begin{array}{ll} \text{portfolio A:} & 0 + Ke^{rT} \\ \text{portfolio B:} & K - S_T + S_T \end{array}$$

  Again, the portfolio A has the higher value.

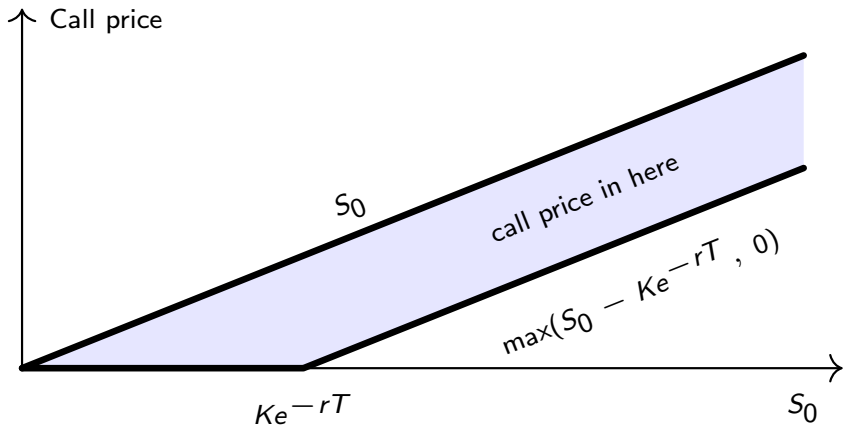
Put-Call Parity for American Options - Left Inequality: case 2 concluded

- Thus, we can conclude that the time-T value of portfolio A is higher than the value of portfolio B.
- $C_0 + K \geq P_0 + S_0 \Longrightarrow C_0 - P_0 \geq S_0 - K$.

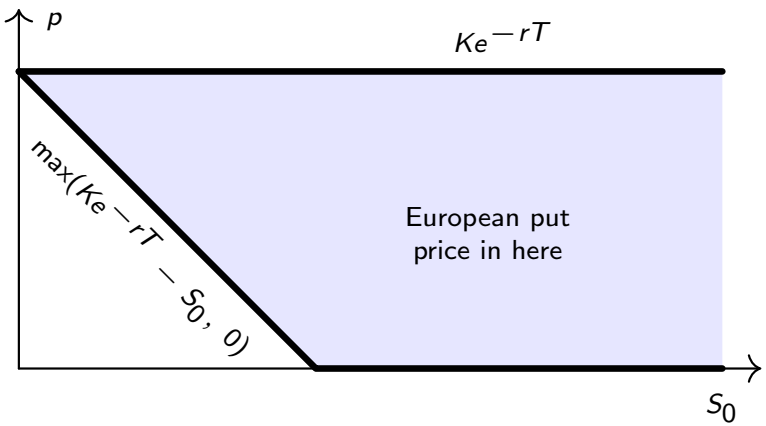
Summary of Bounds for Option Prices

[Non-dividend-paying stock]

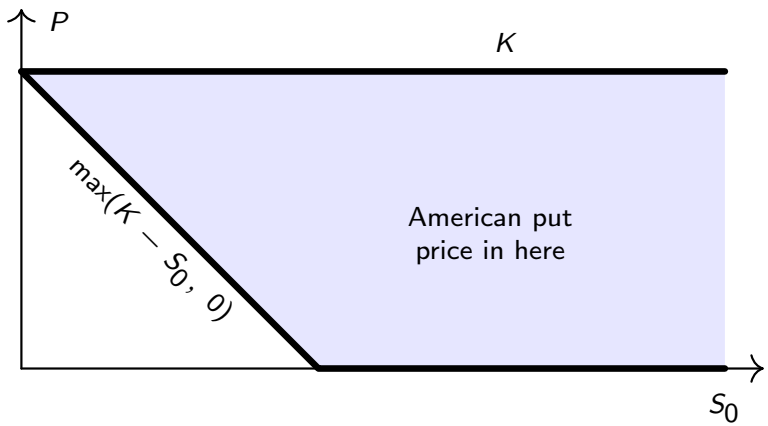
Call (European = American)



European put



American put



Lower/Upper Bounds for Option Prices

Put-Call Parity

II. Options on Dividend-Paying Stocks

Properties of Options on Dividend-Paying Stock

- Recall that the underlying asset's dividend payment affects option prices.
- Hence, the bounds and put-call parity should be modified for options on dividend-paying stock.
- For European options on dividend-paying stocks, we can find the result via a shortcut:
  - Starting from the results for non-dividend-paying stocks, we replace S_0 with the ex-dividend component.

Lower/Upper Bounds and Put-Call Parity for European Options - Discrete Dividends

- Suppose that the underlying assets pay discrete dividends. Let D denote the present value of futures dividends until the option expiration.

- For European call,

$$\max\left(S_0 - D - Ke^{-rT}, 0\right) \leq c_0 \leq S_0 - D$$

- For European put,

$$\max\left(Ke^{-rT} - (S_0 - D), 0\right) \leq p_0 \leq Ke^{-rT}$$

- Put-Call parity,

$$c_0 + Ke^{-rT} = p_0 + S_0 - D$$

Lower/Upper Bounds and Put-Call Parity for European Options - Discrete Dividends: exercise

Q. A European call option on a stock with $K = 20$ and $T = 3$ is priced at \$9. The current stock price is \$30, and the stock is expected to pay dividend of \$2 in $T = 1$ and $T = 2$. The risk-free interest rate is 3%. What is the price of a European put option with the same strike price and expiration date?

Lower/Upper Bounds and Put-Call Parity for European Options - Continuous Dividends

- Suppose that the underlying assets pay continuous dividends with the dividend yield q per annum.

- For European call,

$$\max\left(S_0 e^{-qT} - Ke^{-rT}, 0\right) \leq c_0 \leq S_0 e^{-qT}$$

- For European put,

$$\max\left(Ke^{-rT} - S_0 e^{-qT}, 0\right) \leq p_0 \leq Ke^{-rT}$$

- Put-Call parity,

$$c_0 + Ke^{-rT} = p_0 + S_0 e^{-qT}$$

Summary — Bounds and Parity at a Glance

	Call lower / upper	Put lower / upper
Eur, no div	$\max(S_0 - Ke^{-rT}, 0)$ / S_0	$\max(Ke^{-rT} - S_0, 0)$ / Ke^{-rT}
Eur, disc div	$\max(S_0 - D - Ke^{-rT}, 0)$ / $S_0 - D$	$\max(Ke^{-rT} - (S_0 - D), 0)$ / Ke^{-rT}
Eur, cont div	$\max(S_0 e^{-qT} - Ke^{-rT}, 0)$ / $S_0 e^{-qT}$	$\max(Ke^{-rT} - S_0 e^{-qT}, 0)$ / Ke^{-rT}
Am, no div	= Eur call	lower $= \max(K - S_0, 0)$; upper $= K$

One-line parity. $c_0 - p_0 = (F_0 - K)\, e^{-rT}$

Takeaways.

1. Bounds and parity are *model-free*; they hold whenever the market is frictionless.
2. When they break, suspect frictions: bid-ask, credit risk, short-sale constraints, illiquidity.